

Distributions Maximize entropy under constraints

constraint	interval	$P(x)$ s.t. $\text{Max}(H)$
X	$[a, b]$	①
given μ	$[0, \infty)$	②
given μ, σ^2	$(-\infty, \infty)$	③

~~***~~ (Critical point of Lagrangian) $\Leftrightarrow$ (critical point of H under constraint)

① constraints $\int_a^b P(x) dx = 1$

but why it is maximized on critical point?

$$(\text{Lagrangian}) = - \int_a^b P(x) \ln P(x) dx - \lambda \left(\int_a^b P(x) dx - 1 \right)$$

$p \rightarrow p + \epsilon \mu$ for any μ (variation of calculus)

$$= - \int_a^b (p + \epsilon \mu) \ln(p + \epsilon \mu) dx - \lambda \left(\int_a^b (p + \epsilon \mu) dx - 1 \right)$$

$$\frac{\partial}{\partial \epsilon} (L) = - \int_a^b \mu \ln(p + \epsilon \mu) + (p + \epsilon \mu) \cdot \frac{\mu}{p + \epsilon \mu} dx$$

$$- \lambda \left(\int_a^b \mu dx \right)$$

$\varepsilon \rightarrow 0$ gives.

$$\lim_{\varepsilon \rightarrow 0} \left(\frac{\partial Q}{\partial \varepsilon} \right) = - \int_a^b \nu \ln p + \nu dx - \lambda \int_a^b \nu dx$$
$$= - \int_a^b \nu (\ln p + 1 + \lambda) dx$$

by fundamental lemma of calculus of variation,

$$\ln p + 1 + \lambda = 0 \Rightarrow p(x) = e^{-(\lambda+1)x}$$

by $\int_a^b p(x) dx = 1,$

$$\int_a^b p(x) dx = \int_a^b e^{-(\lambda+1)x} dx = (b-a) \cdot e^{-(\lambda+1)a} = 1$$

$$\Rightarrow \lambda = +\ln(b-a) - 1 = \ln(b-a) - 1$$

$$\therefore p(x) = \frac{1}{b-a}, \quad \underline{X \sim \text{Uniform}(a,b)}$$

(2)

$$\lambda = - \int_0^\infty p(x) \ln p(x) dx - \lambda_0 \left(\int_0^\infty p(x) dx - 1 \right) - \lambda_1 \left(\int_0^\infty x p(x) dx - \mu \right)$$

$$p \rightarrow p(x) + \varepsilon \mu(x)$$

$$\lambda = - \int_0^\infty (p(x) + \varepsilon \mu(x)) \ln(p(x) + \varepsilon \mu(x)) dx$$

$$- \lambda_0 \left(\int_0^\infty p(x) + \varepsilon \mu(x) dx - 1 \right) - \lambda_1 \left(\int_0^\infty x(p(x) + \varepsilon \mu(x)) dx - \mu \right)$$

$$\frac{\partial \lambda}{\partial \varepsilon} = - \int_0^\infty \mu(x) \ln(p(x) + \varepsilon \mu(x)) + \mu(x) dx$$

$$- \lambda_0 \int_0^\infty \mu(x) dx - \lambda_1 \int_0^\infty x \mu(x) dx$$

$$= - \int_0^\infty \mu(x) \left[\ln(p(x) + \varepsilon \mu(x)) + 1 + \lambda_0 + \lambda_1 x \right] dx$$

$\varepsilon \rightarrow 0$ gives,

$$\lim_{\varepsilon \rightarrow 0} \frac{\partial \lambda}{\partial \varepsilon} = - \int_0^\infty \mu(x) \left[\ln(p(x)) + 1 + \lambda_0 + \lambda_1 x \right] dx$$

by Fundamental theorem of calculus,

$$\lambda_1 p(x) + 1 + \lambda_0 + \lambda_1 x = 0$$

$$\Rightarrow p(x) = -\frac{1}{\lambda_1}(x + \lambda_0 + 1)$$

by $\int_0^\infty p(x) dx = 1$ and $\int_0^\infty x p(x) dx = \mu$,

$$1) \int_0^\infty -\frac{1}{\lambda_1}(x + \lambda_0 + 1) dx = -\frac{1}{\lambda_1} \left[\frac{x^2}{2} + (\lambda_0 + 1)x \right]_0^\infty$$

$$= \lim_{x \rightarrow \infty} -\frac{1}{\lambda_1} \left(\frac{x^2}{2} + (\lambda_0 + 1)x \right) = 1$$

$\lim_{x \rightarrow \infty} -\frac{1}{\lambda_1} \left(\frac{x^2}{2} + (\lambda_0 + 1)x \right)$ should be 0, so $\lambda_1 < 0$,

$$-\frac{1}{\lambda_1} = -1 \Rightarrow \lambda_1 = 1$$

$$2) \int_0^\infty x p(x) dx = \int_0^\infty x \left(-\frac{1}{\lambda_1}(x + \lambda_0 + 1) \right) dx$$

$$= -\frac{1}{\lambda_1} \int_0^\infty x(x + \lambda_0 + 1) dx = -\frac{1}{\lambda_1} \left[\frac{x^3}{3} + \frac{(\lambda_0 + 1)x^2}{2} \right]_0^\infty = \mu$$

$$\frac{1}{\lambda_1^2} e^{-(\lambda_0+1)} = + \frac{1}{\lambda_1} = \mu = e^{-(\lambda_0+1)}$$

$$\therefore p(x) = \frac{1}{\mu} e^{-\frac{x}{\mu}}$$

③

$$Z = \frac{- \int_{-\infty}^{\infty} p \ln p \, dx - \lambda_1 \left(\int_{-\infty}^{\infty} x p \, dx - \mu \right)}{I''} - \lambda_2 \left(\int_{-\infty}^{\infty} (x-\mu)^2 p \, dx - \sigma^2 \right) \quad I'$$

since taking all steps on I gives known result,
we may apply those steps on I' and then use
result on I .

$$p \rightarrow p(x) + \epsilon \mu(x)$$

$$= -\lambda_2 \left(\int_{-\infty}^{\infty} (x-\mu)^2 (p(x) + \epsilon \mu(x)) \, dx - \sigma^2 \right)$$

$\frac{\partial}{\partial \epsilon} I'$ gives,

$$= -\lambda_2 \left(\int_{-\infty}^{\infty} (x-\mu)^2 \mu(x) \, dx \right)$$

combining with I gives,

$$\lim_{\epsilon \rightarrow 0} \frac{\partial}{\partial \epsilon} \mathcal{L} = - \int_{-\infty}^{\infty} p(x) \left[\ln p(x) + 1 + \lambda_0 + \lambda_1 x + \lambda_2 (x-\mu)^2 \right] dx$$

by FLVC,

$$\ln p(x) + 1 + \lambda_0 + \lambda_1 x + \lambda_2 (x-\mu)^2 = 0$$

$$\Rightarrow p(x) = \frac{1}{Z} e^{-(\lambda_2 (x-\mu)^2 + \lambda_1 x + \lambda_0 + 1)}$$

by constraints (variance constant should be fixed)

$$(3) \int_{-\infty}^{\infty} (x-\mu)^2 e^{-(\lambda_2 (x-\mu)^2 + \lambda_1 x + \lambda_0 + 1)} dx = \sigma^2$$

substitute $\frac{e^{-(\lambda_0 + 1)}}{2\lambda_2} = A$, then use integration by part.

$$= A \int_{-\infty}^{\infty} -2\lambda_2 (x-\mu) e^{-\lambda_2 (x-\mu)^2} \cdot (x-\mu) e^{-\lambda_1 x} dx =$$

$$= A \left[-e^{-\lambda_2 (x-\mu)^2} (x-\mu) e^{-\lambda_1 x} \right]_{-\infty}^{\infty} - A \int_{-\infty}^{\infty} e^{-\lambda_2 (x-\mu)^2} (-\lambda_1 x + \lambda_1 + 1) e^{-\lambda_1 x} dx$$

should be zero,

$$\lambda_2 > 0$$

$$= + \int_{-\infty}^{\infty} \frac{-\lambda_1 x + \lambda_1 \mu + 1}{2\lambda_2} \frac{e^{-(\lambda_2 (x-\mu)^2 + \lambda_1 x + \lambda_0 \mu)}}{p(x)} dx$$

$$= + \int_{-\infty}^{\infty} \frac{\lambda_1}{2\lambda_2} x p(x) dx + \frac{\lambda_1 \mu}{2\lambda_2} \int_{-\infty}^{\infty} p(x) dx + \frac{1}{2\lambda_2} \int_{-\infty}^{\infty} p(x) dx$$

$$= -\frac{\lambda_1}{2\lambda_2} \mu + \frac{\lambda_1}{2\lambda_2} \mu + \frac{1}{2\lambda_2} = 0$$

$$\therefore \lambda_2 = -\frac{1}{2\sigma^2} < 0$$

$$(2) \int_{-\infty}^{\infty} x p(x) dx = \int_{-\infty}^{\infty} x e^{-\left(\frac{(x-\mu)^2}{2\sigma^2} + \lambda_1 x + \lambda_0 \mu\right)} dx$$

$$\text{let } x - \mu = t, \text{ then } dx = dt.$$

$$= \int_{-\infty}^{\infty} (t + \mu) e^{-\frac{t^2}{2\sigma^2} - \lambda_1 t - \lambda_1 \mu - \lambda_0 \mu} dt \quad \left(\text{let } A = e^{-(\lambda_1 \mu + \lambda_0 \mu)} \right)$$

$$= A \int_{-\infty}^{\infty} (t + \mu) e^{-\frac{t^2 + 2\sigma^2 \lambda_1 t + \sigma^4 \lambda_1^2}{2\sigma^2}} e^{\frac{\sigma^4 \lambda_1^2}{2\sigma^2}} dt$$

$$= A \int_{-\infty}^{\infty} (t+\mu) e^{-\frac{(t+\sigma\lambda_1)^2}{2\sigma^2}} \cdot e^{-\frac{\sigma^2\lambda_1^2}{2}} dt$$

$$= A e^{\frac{\sigma^2\lambda_1^2}{2}} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} \left(\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} t \cdot e^{-\frac{(t+\sigma\lambda_1)^2}{2\sigma^2}} dt + \right.$$

Normal $(-\sigma\lambda_1, \sigma^2)$ expectation

$$\left. \int_{-\infty}^{\infty} \mu \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t+\sigma\lambda_1)^2}{2\sigma^2}} dt \right)$$

$$= A e^{\frac{\sigma^2\lambda_1^2}{2}} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} (\sigma^2\lambda_1 + \mu) = \mu$$

$$= e^{-(\lambda_1\mu + \lambda_0 + 1)} \cdot e^{\frac{\sigma^2\lambda_1^2}{2}} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} (-\sigma^2\lambda_1 + \mu) = \mu$$

to satisfy this condition

$$\text{let } \lambda_1 = 0, \quad e^{-(\lambda_0 + 1)} = \frac{1}{\sqrt{2\pi\sigma^2}}$$

then verify if this value satisfies

$$\int_{-\infty}^{\infty} p(x) dx = 1.$$

$$(1) \int_{-\infty}^{\infty} p(x) dx = \int_{-\infty}^{\infty} e^{-\lambda_2(x-\mu)^2 + \lambda_1(x-\mu_0+1)} dx.$$

$$\lambda_2 = +\frac{1}{2\sigma^2} \quad \lambda_1 = 0 \quad e^{-(\mu_0+1)} = \frac{1}{\sqrt{2\pi\sigma^2}}$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 1$$

pdf of Normal (μ, σ^2)

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

conclusion

Distributions Maximize entropy under constraints

constraint	interval	$p(x)$ Maximize H
X	$[a, b]$	$X \sim \text{Uniform}(a, b),$ $p(x) = \frac{1}{b-a}$
given μ	$[0, \infty)$	$X \sim \text{Exponential}(\mu)$ $p(x) = \frac{1}{\mu} e^{-x/\mu}$
given μ, σ^2	$(-\infty, \infty)$	$X \sim \text{Normal}(\mu, \sigma^2)$ $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$